Spatial Analysis

April 11, 2023

1 Methods

For each sample, we will compare the distribution of each markers in the normal and lesion regions.

- Markers for DD: CD3, CD4, CD8a, CD11c, CD15, CD45, CD68, CD69, CD123, CD279
- Markers for CF: CD1a, CD3, CD4, CD8a, CD11c, CD15, CD45, CD69, CD123, CD279 (CD68 did not pass QC)

Todo

- refine cell segmentation results

2 Results

	Counts				Counts/mm2			
	normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	74	1222	222	5328	631	2874	153	1437
CD15	11	238	7246	2186	93	559	5002	589
CD123	217	1205	3129	9679	1851	2834	2160	2610
CD11c	343	2056	9051	16876	2926	4836	6249	4552
CD68	154	998	3184	10420	1314	2347	2198	2810
area	0.12	0.43	1.45	3.71				

Table 1: DD

	Counts				Counts/mm2			
	normal		lesion		normal		lesion	
	epi	dermis	epi	dermis	epi	dermis	epi	dermis
CD3	422	8435	331	5629	976	4476	819	5574
CD15	16	2644	995	480	37	1403	2462	475
CD123	139	4634	466	1840	321	2459	1153	1822
CD11c	624	8498	1308	4299	1443	4509	3237	4257
area	0.43	1.88	0.40	1.01				

Table 2: CF

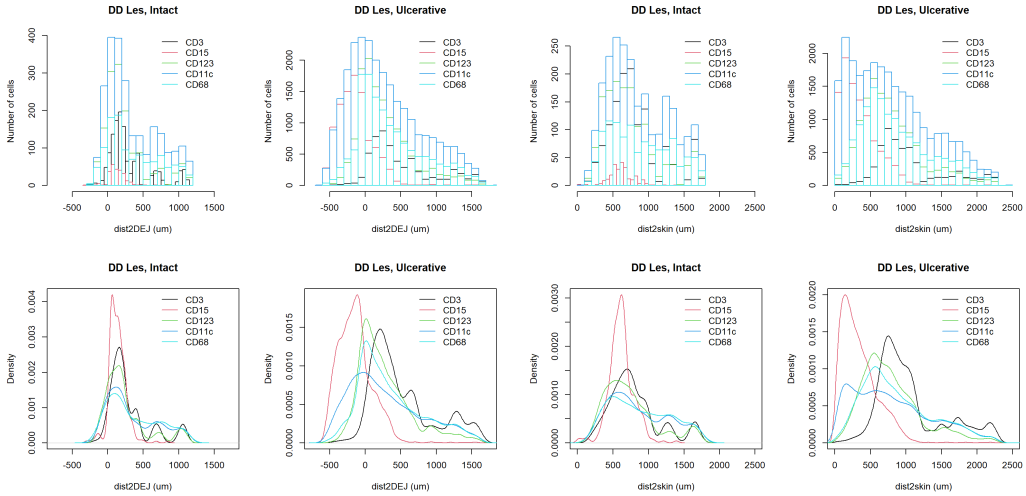


Figure 1: Distance to DEJ (left) and skin surface(right).

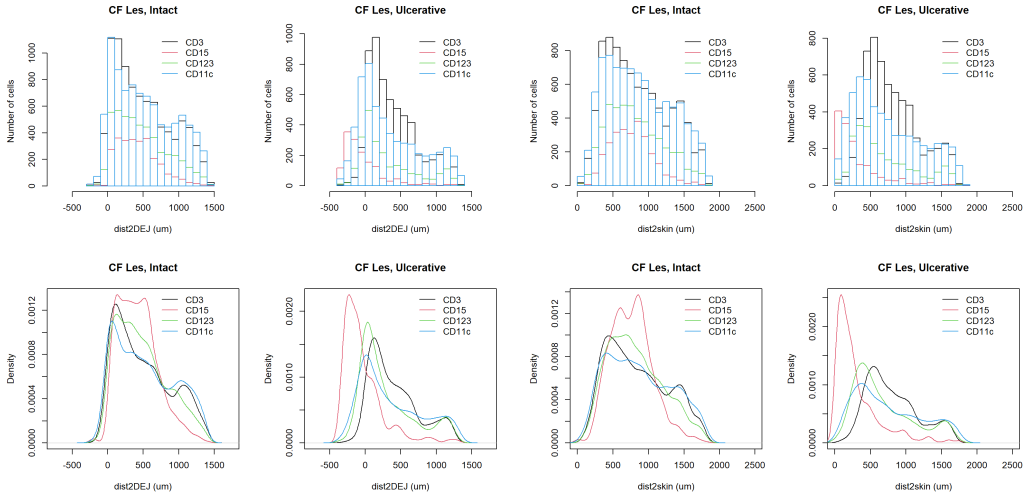


Figure 2: Distance to DEJ (left) and skin surface(right).